NUMERICAL & STATISTICAL METHODS

Subject Code : 23AS4406

UNIT - I

Solution of Algebraic & Transcendental Equations

Syllabus: Introduction - Bisection Method - Iterative method, Regula-falsi method and Newton Raphson method. System of Algebraic equations: Jacobi and Gauss Seidel method.

I. Introduction

Many problems in science & engineering lead to equations like **$f(x) = 0$**. Such equations are of two types :

- **Algebraic equation** : polynomial in x, e.g. $x^3 - 4x + 9 = 0$
- **Transcendental equation** : has trigonometric, exponential or log terms, e.g. $e^x - 3x = 0$, $x - \cos x = 0$

Exact (analytical) solution is not always possible, so we use **numerical methods** which give approximate roots by repeating a process (iteration).

Root of $f(x) = 0$: a number α such that $f(\alpha) = 0$.
Geometrically it is the point where the curve $y = f(x)$ cuts the x-axis.

Intermediate Value Property (starting point)

If $f(x)$ is continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then at least one real root of $f(x) = 0$ lies between a and b.

Steps to locate a root

- Make a table of $f(x)$ for $x = 0, 1, 2, 3 \ldots$ (integer values).
- Find two consecutive values where f changes sign.
- The root lies in that interval, take it as the starting interval.

Example : $f(x) = x^3 - x - 1$. $f(1) = -1$ (negative), $f(2) = 5$ (positive). So a root lies in $(1, 2)$.

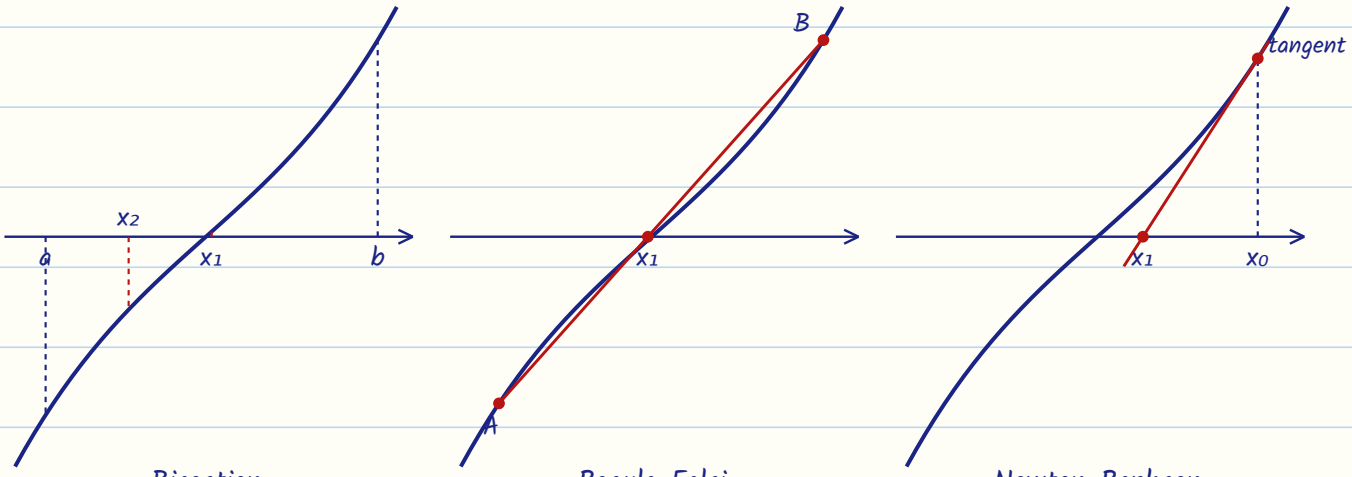
Types of methods

- **Bracketing methods** (need interval) : Bisection, Regula-Falsi
- **Open / iterative methods** (need initial guess) : Iteration, Newton-Raphson

Rate of convergence (order of a method)

If e_n = error at nth step and $e_{n+1} = C \cdot e_n^p$ then p is the **order of convergence**.

Method	Order p	Needs derivative?	Type
Bisection	1 (slow)	No	Bracketing
Regula-Falsi	1 (about 1.0)	No	Bracketing
Iteration	1 (linear)	Needs $\varphi'(x) < 1$	Open
Newton-Raphson	2 (quadratic)	Yes, $f'(x)$	Open



2. Bisection Method

Idea : repeatedly halve the interval containing the root (also called Bolzano's / interval-halving method).

Algorithm

- Choose a, b such that $f(a) \cdot f(b) < 0$.
- Find the mid-point $x_1 = (a + b) / 2$.
- If $f(x_1) = 0$, x_1 is the root. Stop.
- If $f(a) \cdot f(x_1) < 0$, root is in (a, x_1), so put $b = x_1$. Otherwise root is in (x_1, b), so put $a = x_1$.
- Repeat till $|b - a|$ is less than the required accuracy.

Each step halves the interval, so after n steps the length is $(b - a)/2^n$. Number of iterations needed for error ε : $n \geq \log(b - a) - \log \varepsilon / \log 2$

Example 1

Find a root of $x^3 - x - 1 = 0$ by bisection (correct to 2 decimals).

Here $f(1) = -1 < 0$ and $f(2) = 5 > 0$, root in $(1, 2)$.

n	a	b	x = (a+b)/2	f(x)
1	1	2	1.5	+ 0.875
2	1	1.5	1.25	− 0.2969
3	1.25	1.5	1.375	+ 0.2246
4	1.25	1.375	1.3125	− 0.0515
5	1.3125	1.375	1.34375	+ 0.0826
6	1.3125	1.34375	1.3281	+ 0.0146
7	1.3125	1.3281	1.3203	− 0.0187

Root lies between 1.3203 and 1.3281, so the root ≈ **1.32** (exact root = 1.3247).

Merits and demerits

- Merit : always converges, very simple, no derivative needed.
- Demerit : converges slowly (linear), needs many iterations, cannot find double (even multiplicity) roots.

Example 2 (to practise)

Find the positive root of $x^3 - 4x - 9 = 0$ using bisection. Take $f(2) = -9$, $f(3) = 6$. First approximation $x_1 = 2.5$, $f(2.5) = -3.375$ (negative) so root in $(2.5, 3)$. Continue ... root ≈ 2.706.

3. Method of Iteration (Fixed Point)

Write $f(x) = 0$ in the form $x = \varphi(x)$. Start with x_0 and generate

$$x_{n+1} = \varphi(x_n), \quad n = 0, 1, 2, \ldots$$

Condition for convergence : $|\varphi'(x)| < 1$ for all x in the interval containing the root. Smaller the value of $|\varphi'|$, faster the convergence.

Procedure

- Locate the root in (a, b) by sign change.
- Rewrite $f(x) = 0$ as $x = \varphi(x)$ such that $|\varphi'(x)| < 1$ in (a, b).
- Take $x_0 = a$ or b or mid-point, find $x_1 = \varphi(x_0)$, $x_2 = \varphi(x_1)$ ... till two successive values agree to required decimals.

Example 3

Find the root of $\cos x - 3x + 1 = 0$ by iteration (correct to 3 decimals).

$f(0) = 2 > 0$, $f(1) = 0.54 - 2 = -1.46 < 0$, so root in $(0, 1)$.

Write $x = (1 + \cos x)/3 = \varphi(x)$. Then $\varphi'(x) = -\sin x / 3$, $|\varphi'| \leq 1/3 < 1$, so the method converges.

Take $x_0 = 0.5$ (radians).

n	x_n	$x_{n+1} = (1 + \cos x_n)/3$
0	0.5	0.62586
1	0.62586	0.60349
2	0.60349	0.60779
3	0.60779	0.60697
4	0.60697	0.60713
5	0.60713	0.60710

Hence the root is $x \approx 0.607$ (correct to 3 decimals).

Note

If $|\varphi'(x)| > 1$ the iterates move away from root, so choose a different form of $\varphi(x)$. For $x^3 + x^2 - 1 = 0$ one may use $x = 1/\sqrt{(1 + x)}$ (since $|\varphi'| < 1$ near the root $x \approx 0.7549$).

4. Regula-Falsi Method (False Position)

Idea : replace the curve by a chord (straight line) joining the points $A(a, f(a))$ and $B(b, f(b))$. The point where the chord cuts the x-axis is the next approximation.

Equation of chord AB : $(y - f(a)) / (f(b) - f(a)) = (x - a)/(b - a)$. Put $y = 0$:

$$x_1 = [\, a \cdot f(b) - b \cdot f(a) \,] / [\, f(b) - f(a) \,]$$

Algorithm

- Take a, b with $f(a) \cdot f(b) < 0$.
- Compute x_1 by the formula above.

- If $f(a) \cdot f(x_1) < 0$ then $b = x_1$, else $a = x_1$.
- Repeat with the new interval till required accuracy.

Example 4

Find a real root of $x^3 - 2x - 5 = 0$ by Regula-Falsi.

$f(2) = -1$ (negative), $f(3) = 16$ (positive). Root in (2, 3). $a = 2$, $b = 3$.

Step 1 : $x_1 = [2(16) - 3(-1)] / [16 + 1] = 35/17 = 2.05882$, $f(x_1) = -0.3908$ (negative) so $a = 2.05882$.

Step 2 : $x_2 = 2.08126$, $f(x_2) = -0.1472$ (negative), $a = 2.08126$.

Step 3 : $x_3 = 2.08964$, $f = -0.0547$.

Step 4 : $x_4 = 2.09274$, $f = -0.0202$.

Step 5 : $x_5 = 2.09388$, $f = -0.0075$.

Values are slowly approaching the root **2.0946**. (Correct to 2 decimals : 2.09)

Comparison

- Bisection uses only the mid-point; Regula-Falsi uses the function values so it is usually faster.
- But one end point often stays fixed, so the convergence is only linear.

5. Newton-Raphson Method

Idea : replace the curve near the root by its tangent at x_0. Tangent at $(x_0, f(x_0))$: $y - f(x_0) = f'(x_0)(x - x_0)$. It cuts x-axis ($y = 0$)

at x_1.

$$x_{n+1} = x_n - f(x_n) / f'(x_n) \ , \ n = 0, 1, 2 \ ...$$

Algorithm

- Choose initial guess x_0 close to root (usually the end where $f(x) \cdot f''(x) > 0$).
- Compute $f(x_n)$ and $f'(x_n)$, find x_{n+1}.
- Stop when $|x_{n+1} - x_n| < \varepsilon$.

Convergence condition : $|f(x) \cdot f''(x)| < |f'(x)|^2$ in the neighbourhood of root. Order of convergence is **2** (quadratic), i.e. number of correct decimal places nearly doubles every step.

Example 5 : square root formula

Find $\sqrt{12}$: this is the root of $x^2 - 12 = 0$. Here $f' = 2x$ so

$x_{n+1} = x_n - (x_n^2 - 12)/(2x_n) = \frac{1}{2} (x_n + 12/x_n)$

Take $x_0 = 3$: $x_1 = \frac{1}{2}(3 + 4) = 3.5$; $x_2 = \frac{1}{2}(3.5 + 3.4286) = 3.46429$; $x_3 = 3.46410$. So $\sqrt{12} =$ **3.4641**.

Example 6

Find the root of $x^3 - 2x - 5 = 0$ near 2 by Newton-Raphson.

$f(x) = x^3 - 2x - 5$, $f'(x) = 3x^2 - 2$. $x_0 = 2$:

$x_1 = 2 - (-1)/10 = 2.1$

$x_2 = 2.1 - f(2.1)/f'(2.1) = 2.1 - (0.061/11.23) = 2.09457$

$x_3 = 2.09455$, so root = **2.0946**

Compare : Regula-Falsi needed more than 5 steps to reach the same accuracy.

Special formulae from Newton-Raphson

To find	Formula (x_{n+1})
Reciprocal 1/N	$x_n(2 - N\,x_n)$
Square root $\sqrt{N}$	$\frac{1}{2}\,(\,x_n + N/x_n\,)$
kth root of N	$[\,(k-1)\,x_n + N/x_n^{(k-1)}\,]\,/\,k$
Reciprocal sq. root $1/\sqrt{N}$	$\frac{1}{2}\,x_n\,(3 - N\,x_n^2)$

Demerits

- Fails if $f'(x) = 0$ near the root; needs derivative every step; wrong x_0 may diverge.

6. System of Linear Equations : Iterative Methods

Direct methods (Gauss elimination) give exact answers after fixed steps. For large systems with many zeros, **iterative methods** are better. Let the system be

$a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$, $a_3x + b_3y + c_3z = d_3$

Condition (diagonally dominant) : in every equation the coefficient of the leading unknown is larger in magnitude than the sum of the other coefficients : $|a_1| > |b_1| + |c_1|$, $|b_2| > |a_2| + |c_2|$, $|c_3| > |a_3| + |b_3|$. This guarantees convergence.

Rewrite as : $x = (d_1 - b_1y - c_1z)/a_1$, $y = (d_2 - a_2x - c_2z)/b_2$, $z = (d_3 - a_3x - b_3y)/c_3$

6.1 Jacobi's Method

Start with $x_0 = y_0 = z_0 = 0$ (initial values). Compute all new values **using only old values** of previous iteration:

$$x^{(k+1)} = (d_1 - b_1y^{(k)} - c_1z^{(k)})/a_1 \text{ , similarly } y, z$$

6.2 Gauss-Seidel Method

Same equations, but **use the latest available value immediately** : as soon as $x^{(k+1)}$ is found, it is used in the y-equation, and $x^{(k+1)}$, $y^{(k+1)}$ in the z-equation.

Example 7

Solve $10x + y + z = 12$, $2x + 10y + z = 13$, $2x + 2y + 10z = 14$.

Diagonally dominant ($10 > 2$, $10 > 3$, $10 > 4$). So $x = (12 - y - z)/10$, $y = (13 - 2x - z)/10$, $z = (14 - 2x - 2y)/10$.

Jacobi (start 0, 0, 0)

Iteration	x	y	z
1	1.2	1.3	1.4
2	0.93	0.92	0.90
3	1.018	1.024	1.030
4	0.9946	0.9934	0.9916

Tends to $x = 1$, $y = 1$, $z = 1$.

Gauss-Seidel (start 0, 0, 0)

Iteration	x	y	z
1	1.2	1.06	0.948
2	0.9992	1.0054	0.9991
3	0.9996	1.0002	1.0001
4	1.0000	1.0000	1.0000

Gauss-Seidel reaches the answer in about half the iterations.

Gauss-Seidel converges **about twice as fast** as Jacobi (when both converge) and needs less storage since new values overwrite old ones.

Jacobi vs Gauss-Seidel

Jacobi	Gauss-Seidel
Uses old values only	Uses latest values
Slower convergence	Faster (twice)
Needs 2 sets of storage	One set only
Also called method of simultaneous displacement	Method of successive displacements

7. Important Points & Quick Revision

- Bisection : $x_{n+1} = (a + b)/2$, always converges, order 1.
- Iteration : $x = \varphi(x)$, needs $|\varphi'(x)| < 1$, order 1.
- Regula-Falsi : $x = [a\ f(b) - b\ f(a)] / [f(b) - f(a)]$, order ≈ 1.
- Newton-Raphson : $x = x - f/f'$, order 2, fastest.
- Jacobi / Gauss-Seidel : need diagonal dominance, G-S twice faster.

Typical exam questions

- Find a real root of $x^3 - 4x - 9 = 0$ by bisection (answer 2.706).
- Find the root of $x\,e^x = 2$ by Newton-Raphson / Regula-Falsi (root ≈ 0.8526).
- Solve $27x + 6y - z = 85$, $6x + 15y + 2z = 72$, $x + y + 54z = 110$ by Gauss-Seidel ($x = 2.43$, $y = 3.57$, $z = 1.92$).
- Derive Newton-Raphson formula. Find $\sqrt{N}$ and $1/N$ using it.

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UNIT - II

Interpolation & Curve Fitting

Syllabus: Finite differences - Newton's forward and backward interpolation formulae - Lagrange's formula. Curve fitting: Fitting of straight line, second-degree and Exponential curve by method of least squares.

I. Introduction to Interpolation

Interpolation is the process of finding the value of a function $y = f(x)$ for a value of x lying **inside** the range of given data (x_0, y_0), (x_1, y_1), ... (x_n, y_n). Finding a value outside the range is called **extrapolation**.

We replace the unknown function by a simple polynomial $P(x)$ of degree $\leq n$ which passes through all the $(n + 1)$ given points : $P(x_i) = y_i$, $i = 0, 1, \ldots n$.

- **Equal spacing** of x ($x_1 - x_0 = h$) : Newton's forward / backward formulae.
- **Unequal spacing** : Lagrange's formula.

2. Finite Differences

Let the values of x be equally spaced with step h : $x_i = x_0 + ih$, and $y_i = f(x_i)$.

2.1 Forward difference operator Δ

$$\Delta y_i = y_{i+1} - y_i \text{ (first forward difference)}$$

$$\Delta^2 y_i = \Delta y_{i+1} - \Delta y_i \ , \ \Delta^n y_i = \Delta^{n-1} y_{i+1} - \Delta^{n-1} y_i$$

2.2 Backward difference operator ∇

$$\nabla y_i = y_i - y_{i-1} \ , \ \nabla^2 y_i = \nabla y_i - \nabla y_{i-1}$$

2.3 Other operators

- **Shift operator E** : $E\, f(x) = f(x + h)$, so $E\, y_i = y_{i+1}$ and $E^n\, y_i = y_{i+n}$.
- **Central difference δ** : $\delta\, y_i = y(i+½) - y(i-½)$.
- **Averaging μ** : $\mu\, y_i = ½\, [\, y(i+½) + y(i-½)\,]$.

Relations : $\Delta = E - 1$, $\nabla = 1 - E^{-1}$, $E = 1 + \Delta$, $\Delta\nabla = \nabla\Delta = \delta^2$, $\nabla = \Delta E^{-1}$

Important results

- The nth difference of a polynomial of degree n is **constant** ($= n!\, h^n\, a_n$) and the (n+1)th difference is zero.
- $\Delta(c) = 0$; $\Delta(e^{ax}) = (e^{ah} - 1)\, e^{ax}$; $\Delta(f + g) = \Delta f + \Delta g$.

2.4 Forward difference table (n = 4 points shown)

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_0	y_0	Δy_0	$\Delta^2 y_0$	$\Delta^3 y_0$	$\Delta^4 y_0$
x_1	y_1	Δy_1	$\Delta^2 y_1$	$\Delta^3 y_1$	
x_2	y_2	Δy_2	$\Delta^2 y_2$		
x_3	y_3	Δy_3			
x_4	y_4				

In backward table the same entries are written along the **last row** :

∇y_n, $\nabla^2 y_n$, $\nabla^3 y_n$ ...

Example 1 : difference table for $y = x^3$

x	$y = x^3$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1	1	7	12	6	0
2	8	19	18	6	
3	27	37	24		
4	64	61			
5	125				

Third difference is constant ($6 = 3!$) so the data come from a cubic polynomial.

Missing term problem

If one value is missing among n known values, assume the (n)th differences are zero, i.e. $\Delta^n y_0 = 0$; expand in terms of E : $(E - 1)^n y_0 = 0$.

Example : 4 values known, 1 missing → use $\Delta^4 y_0 = 0$ → $y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$.

3. Newton's Forward Interpolation Formula

Used when the required x is **near the beginning** of the table. Put **$p = (x - x_0)/h$**.

$$y(x) = y_0 + p\Delta y_0 + [p(p-1)/2!]\,\Delta^2 y_0 + [p(p-1)(p-2)/3!]\,\Delta^3 y_0 + \dots$$

Derivation idea : $y(x_0 + ph) = E^p\, y_0 = (1 + \Delta)^p\, y_0$ and expand by binomial theorem.

Example 2

From the table of $y = x^3$ (Example 1) find $y(1.5)$.

$x_0 = 1$, $h = 1$, $p = (1.5 - 1)/1 = 0.5$. $y_0 = 1$, $\Delta y_0 = 7$, $\Delta^2 y_0 = 12$, $\Delta^3 y_0 = 6$.

$y(1.5) = 1 + (0.5)(7) + [(0.5)(-0.5)/2](12) + [(0.5)(-0.5)(-1.5)/6](6)$

$= 1 + 3.5 - 1.5 + 0.375 =$ **3.375** (exact value $1.5^3 = 3.375$ ✓)

4. Newton's Backward Interpolation Formula

Used when x is **near the end** of the table. Put **$p = (x - x_n)/h$** (p is negative).

$$y(x) = y_n + p\nabla y_n + [p(p+1)/2!]\,\nabla^2 y_n + [p(p+1)(p+2)/3!]\,\nabla^3 y_n + \dots$$

Example 3

Find $y(4.5)$ from the same table.

$x_n = 5$, $h = 1$, $p = (4.5 - 5)/1 = -0.5$. Last row values : $y_n = 125$, $\nabla y_n = 61$, $\nabla^2 y_n = 24$, $\nabla^3 y_n = 6$.

$y(4.5) = 125 + (-0.5)(61) + [(-0.5)(0.5)/2](24) + [(-0.5)(0.5)(1.5)/6](6)$

$= 125 - 30.5 - 3 - 0.375 = \mathbf{91.125}$ (exact $4.5^3 = 91.125$ ✓)

Rule : x near start → Newton's forward ; x near end → Newton's backward ; x in the middle → central difference formulae (Gauss, Stirling, Bessel).

Example 4 (population problem)

Table of population of a town (in thousands) :

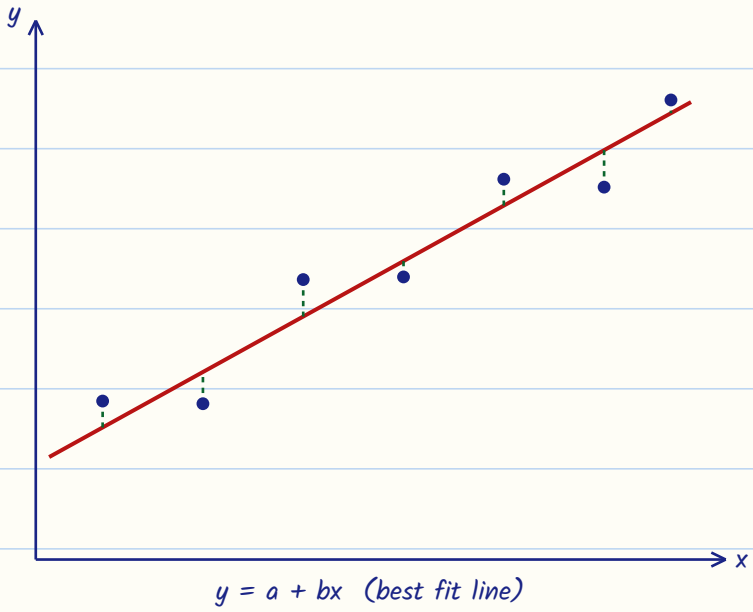
Year	1931	1941	1951	1961	1971
Population	40	60	79	103	133

Differences : $\Delta = 20, 19, 24, 30$; $\Delta^2 = -1, 5, 6$; $\Delta^3 = 6, 1$; $\Delta^4 = -5$.

Estimate the population in 1936 : $p = (1936 - 1931)/10 = 0.5$

$y = 40 + 0.5(20) + [0.5(-0.5)/2](-1) + [0.5(-0.5)(-1.5)/6](6) + [0.5(-0.5)(-1.5)(-2.5)/24](-5)$

$= 40 + 10 + 0.125 + 0.375 + 0.1953 \approx \mathbf{50.70}$ **thousand**



Principle of least squares:
minimise $E = \Sigma\, d_i^2$
where $d_i = y_i - (a + bx_i)$
(green dashed = residuals)
sum of squares of errors should be the least

5. Lagrange's Interpolation Formula

For **unequally spaced** x values (also works for equal spacing). Given (x_0,y_0), (x_1,y_1) ... (x_n,y_n) :

$$y(x) = \Sigma\, y_i \cdot L_i(x)\ ,\ L_i(x) = \Pi\ (x - x_j)/(x_i - x_j)\ (j \neq i)$$

For three points (x_0, x_1, x_2):

$$y = (x-x_1)(x-x_2)/[(x_0-x_1)(x_0-x_2)]\ y_0 + (x-x_0)(x-x_2)/[(x_1-x_0)(x_1-x_2)]\ y_1 + (x-x_0)(x-x_1)/[(x_2-x_0)(x_2-x_1)]\ y_2$$

Properties

- Polynomial has degree n for (n + 1) points and is unique.
- $L_i(x_i) = 1$ and $L_i(x_j) = 0$ for $i \neq j$.
- No need of difference table, but adding a new point means recomputing everything.

Example 5

Using Lagrange's formula, find y(3) given (0, 2), (1, 3), (2, 12), (5, 147).

$L_0 = (3-1)(3-2)(3-5) / [(0-1)(0-2)(0-5)] = (2)(1)(-2)/(-10) = 0.4$

$L_1 = (3-0)(3-2)(3-5) / [(1-0)(1-2)(1-5)] = (3)(1)(-2)/(4) = -1.5$

$L_2 = (3-0)(3-1)(3-5) / [(2-0)(2-1)(2-5)] = (3)(2)(-2)/(-6) = 2$

$L_3 = (3-0)(3-1)(3-2) / [(5-0)(5-1)(5-2)] = 6/60 = 0.1$

$y(3) = 2(0.4) + 3(-1.5) + 12(2) + 147(0.1) = 0.8 - 4.5 + 24 + 14.7 = \mathbf{35}$

The polynomial is $y = x^3 + x^2 - x + 2$, check : $27 + 9 - 3 + 2 = 35$ ✓

Inverse interpolation

To find x for a given y, interchange the roles : apply Lagrange's formula with x and y swapped, $x = \Sigma\, x_i \cdot \Pi\, (y - y_j)/(y_i - y_j)$.

Newton vs Lagrange

Newton forward / backward	Lagrange
Equal intervals only	Equal or unequal intervals
Uses difference table	Direct, no table
Easy to add more terms	Must redo whole work
Less calculation	More calculation

6. Curve Fitting : Method of Least Squares

Given data (x_i, y_i), $i = 1 \ldots n$, we find a curve $y = f(x)$ that **best fits** the data (need not pass through every point).

Principle of least squares : choose the constants so that the sum of squares of errors $E = \Sigma\, [\, y_i - f(x_i)\,]^2$ is minimum. (Put $\partial E/\partial a = 0$, $\partial E/\partial b = 0$ to get **normal equations**.)

6.1 Straight line $y = a + bx$

Normal equations :

$$\Sigma y = n\, a + b\, \Sigma x$$

$$\Sigma xy = a\,\Sigma x + b\,\Sigma x^2$$

Solve for a and b.

Example 6

Fit a straight line to : x = 1,2,3,4,5 ; y = 14,27,40,55,68.

x	y	xy	x^2
1	14	14	1
2	27	54	4
3	40	120	9
4	55	220	16
5	68	340	25
15	204	748	55

204 = 5a + 15b ; 748 = 15a + 55b. Multiply first by 3 : 612 = 15a + 45b. Subtract : 136 = 10b → **b = 13.6**, a = (204 − 204)/15 = 0.

So the line is **y = 13.6 x**.

6.2 Second degree parabola $y = a + bx + cx^2$

$$\Sigma y = na + b\Sigma x + c\Sigma x^2$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$$

$$\Sigma x^2y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$$

Example 7

Fit $y = a + bx + cx^2$ to : x = 0,1,2,3,4 ; y = 1, 1.8, 1.3, 2.5, 6.3.

Sums : n = 5, $\Sigma x = 10$, $\Sigma x^2 = 30$, $\Sigma x^3 = 100$, $\Sigma x^4 = 354$, $\Sigma y = 12.9$, $\Sigma xy = 37.1$, $\Sigma x^2y = 130.3$.

12.9 = 5a + 10b + 30c ; 37.1 = 10a + 30b + 100c ; 130.3 = 30a + 100b + 354c

Solving : **a = 1.42 , b = −1.07 , c = 0.55**. So $y = 1.42 - 1.07x + 0.55x^2$.

6.3 Exponential curve $y = a\,e^{bx}$

Take logs (base 10) : $\log y = \log a + (b \log e)\,x$, i.e. **$Y = A + Bx$**

where $Y = \log y$, $A = \log a$, $B = b \log e$. It becomes a straight line fit in (x, Y).

Normal equations : $\Sigma Y = nA + B\Sigma x$, $\Sigma xY = A\Sigma x + B\Sigma x^2$. Then $a = 10^A$, $b = B / \log e$ ($\log e = 0.4343$).

Example 8

Fit $y = a\,e^{bx}$: x = 1,2,3,4,5 ; y = 2.0, 4.5, 10.1, 22.8, 51.2.

$Y = \log y$ = 0.3010, 0.6532, 1.0043, 1.3579, 1.7093. $\Sigma x = 15$, $\Sigma x^2 = 55$, $\Sigma Y = 5.0258$, $\Sigma xY = 18.5985$.

$5.0258 = 5A + 15B$; $18.5985 = 15A + 55B \rightarrow$ **B = 0.3521 , A = −0.0512**

a = 10^(−0.0512) = 0.889 , b = 0.3521/0.4343 = 0.811. Curve : **y = 0.889 e^(0.811 x)**

Similarly $y = a\,x^b$ (power curve) ...

Take log : $\log y = \log a + b \log x$, so $Y = A + bX$ with $X = \log x$, $Y = \log y$. Fit a line in (X, Y).

6A. More Solved Examples

Example 9 : Newton forward with non-polynomial data

Given sin 45° = 0.7071, sin 50° = 0.7660, sin 55° = 0.8192, sin 60° = 0.8660. Find sin 52°.

x°	y = sin x	Δy	Δ²y	Δ³y
45	0.7071	0.0589	−0.0057	−0.0007
50	0.7660	0.0532	−0.0064	
55	0.8192	0.0468		
60	0.8660			

$x_0 = 45$, $h = 5$, $p = (52 - 45)/5 = 1.4$.

$y = 0.7071 + 1.4(0.0589) + [1.4(0.4)/2](-0.0057) + [1.4(0.4)(-0.6)/6](-0.0007)$

$= 0.7071 + 0.08246 - 0.00160 + 0.00004 =$ **0.7880** (actual sin 52° = 0.7880 ✓)

Example 10 : finding the missing value

Find the missing term : x = 0,1,2,3,4 ; y = 1, 3, 9, –, 81.

Four values are known so $\Delta^4 y_0 = 0$: $y_4 - 4y_3 + 6y_2 - 4y_1 + y_0 = 0$

$81 - 4y_3 + 54 - 12 + 1 = 0 \rightarrow 124 = 4y_3 \rightarrow$ **$y_3 = 31$**.

Example 11 : Lagrange with three points

Find y(2) from (1, 1), (3, 27), (4, 64) (these are cubes).

$y(2) = [(2-3)(2-4)]/[(1-3)(1-4)] \cdot 1 + [(2-1)(2-4)]/[(3-1)(3-4)] \cdot 27 + [(2-1)(2-3)]/[(4-1)(4-3)] \cdot 64$

$= (2/6)(1) + (-2/-2)(27) + (-1/3)(64) = 0.3333 + 27 - 21.3333 =$ **6.0**

(Cube of 2 is 8, the difference arises because we used only a quadratic through three points.)

Error in interpolation

If $y = f(x)$ is approximated by a polynomial $P_n(x)$ through $(n + 1)$ points then

$$f(x) - P_n(x) = [\,(x-x_0)(x-x_1)\ldots(x-x_n) / (n+1)!\,] \cdot f^{(n+1)}(\xi)$$

where ξ is some point with $x_0 < \xi < x_n$.

Error is small when x is near the middle of the data points and h is small.

Example 12 : straight line with intercept

Fit $y = a + bx$: $x = 0, 1, 2, 3, 4$; $y = 1, 3, 4, 6, 8\ldots$ $(n = 5)$

$\Sigma x = 10$, $\Sigma y = 22$, $\Sigma xy = 0 + 3 + 8 + 18 + 32 = 61$, $\Sigma x^2 = 30$.

$22 = 5a + 10b$; $61 = 10a + 30b$. From first : $44 = 10a + 20b$.

Subtract : $17 = 10b \rightarrow b = 1.7$, then $a = (22 - 17)/5 = 1.0$.

Best line : **$y = 1 + 1.7\,x$**. Estimate at $x = 5$: $y = 9.5$.

Example 13 : power curve $y = a\,x^b$

$x = 1, 2, 3, 4$; $y = 2.0, 5.7, 10.4, 16.0$ (roughly $y = 2x^{1.5}$).

$X = \log x = 0, 0.3010, 0.4771, 0.6021$; $Y = \log y = 0.3010, 0.7559, 1.0170, 1.2041$.

$\Sigma X = 1.3802$, $\Sigma Y = 3.2780$, $\Sigma XY = 1.4377$ (approx), $\Sigma X^2 = 0.6807$ (approx)

Normal equations on $Y = A + bX$ give $b \approx 1.5$, $A \approx 0.30 \rightarrow a = 10^{0.30} = 2.0$. So $y \approx 2.0\,x^{1.5}$.

Remarks on curve fitting

- Number of normal equations = number of constants in the curve.
- To reduce the size of calculation, shift origin : $u = (x - A)/h$ when x values are equally spaced.
- Goodness of fit : compare Σ of squared residuals of different curves, least value is best.

7. Quick Revision

- $\Delta = E - 1$, $\nabla = 1 - E^{-1}$, $p = (x - x_0)/h$ for forward, $p = (x - x_n)/h$ for backward.
- Forward : near top of table ; Backward : near bottom ; Lagrange : unequal gaps.
- Least squares : straight line (2 normal eqs), parabola (3 normal eqs), exponential (take log first).

Exam questions

- Derive Newton's forward interpolation formula.
- Using Lagrange's formula, find y(10) given (5,12), (6,13), (9,14), (11,16) → y(10) = 14.67.
- Fit $y = ab^x$ or $y = a\,e^{(bx)}$ to the given data by least squares.
- Prove that $\Delta = E - 1$ and $\nabla = 1 - E^{-1}$; find the missing term in a table.

NUMERICAL & STATISTICAL METHODS

Subject Code : 23AS4406

UNIT - III

Solution of Initial Value Problems to Ordinary Differential Equation

Syllabus: Numerical solution of ODEs: Solution by Taylor's series - Picard's Method of successive approximations - Euler's and modified Euler's methods - Runge-Kutta methods (second and fourth order).

I. Introduction

Many problems in engineering lead to a **differential equation** whose analytical solution is difficult. Numerical methods give the value of y at chosen points of x.

Initial Value Problem (IVP) : $dy/dx = f(x, y)$ with $y(x_0) = y_0$.
We want y at $x_1 = x_0 + h$, $x_2 = x_0 + 2h$, ... (h = step size).

Two kinds of methods

- **Series / analytical-type** : Taylor series, Picard's method (give y as a function of x).
- **Step-by-step (marching) methods** : Euler, modified Euler, Runge-Kutta (give y at a table of x values).

Sources of error

- **Truncation error** : due to cutting the series / formula after finite terms.

- **Round-off error** : due to limited number of decimal places.
- Smaller h → less truncation error but more steps and more round-off.

2. Taylor's Series Method

Let $y = y(x)$ be the solution of $dy/dx = f(x, y)$, $y(x_0) = y_0$. Expand $y(x)$ about x_0 :

$$y(x) = y_0 + (x-x_0)\,y_0' + (x-x_0)^2/2!\;y_0'' + (x-x_0)^3/3!\;y_0''' + \dots$$

Procedure

- Get $y_0' = f(x_0, y_0)$ from the given equation.
- Differentiate the equation again and again to find y'', y''', $y^{(4)}$ at (x_0, y_0).
- Substitute in the series. Use few terms (up to h^4 is common).
- For the next point, take the new point as starting point (x_1, y_1) and repeat.

Example 1

Solve $dy/dx = x + y$, $y(0) = 1$ and find $y(0.1)$.

$y' = x + y \rightarrow y'(0) = 1$. $y'' = 1 + y' \rightarrow y''(0) = 2$. $y''' = y'' \rightarrow 2$. $y^{(4)} = 2$.

$y = 1 + x(1) + x^2/2\,(2) + x^3/6\,(2) + x^4/24\,(2) = 1 + x + x^2 + x^3/3 + x^4/12$

At $x = 0.1$: $y = 1 + 0.1 + 0.01 + 0.000333 + 0.0000083 =$ **1.11034**

(exact solution $y = 2e^x - x - 1$ gives 1.110342 ✓)

Example 2

Solve $dy/dx = x + y^2$, $y(0) = 1$. Find $y(0.1)$ and $y(0.2)$.

$y' = x + y^2 \to 1$; $y'' = 1 + 2yy' \to 1 + 2 = 3$; $y''' = 2y'^2 + 2yy'' \to 2 + 6 = 8$; $y^{(4)} = 6y'y'' + 2yy''' \to 18 + 16 = 34$

$y = 1 + x + 3x^2/2 + 8x^3/6 + 34x^4/24 + \ldots$

$y(0.1) = 1 + 0.1 + 0.015 + 0.00133 + 0.00014 = \mathbf{1.1165}$

$y(0.2) = 1 + 0.2 + 0.06 + 0.01067 + 0.00227 = \mathbf{1.2729}$

Merits & demerits of Taylor's method

- Merit : gives a continuous solution, accuracy can be improved by adding terms.
- Demerit : needs higher derivatives of $f(x, y)$, which is tedious for complicated functions.

3. Picard's Method of Successive Approximations

Integrate $dy/dx = f(x, y)$ from x_0 to x : $y = y_0 + \int f(x, y)\, dx$ (limits x_0 to x). Start with y_0 and improve:

$$y^{(1)} = y_0 + \int f(x, y_0)\, dx \ , \ y^{(2)} = y_0 + \int f(x, y^{(1)})\, dx$$

$$\boxed{y^{(n)} = y_0 + \int f(x, y^{(n-1)})\, dx}$$

Continue until two successive approximations agree to the required accuracy.

Example 3

Solve $dy/dx = x + y$, $y(0) = 1$ by Picard's method. Find $y(0.1)$.

$f(x, y) = x + y$, $x_0 = 0$, $y_0 = 1$.

$y^{(1)} = 1 + \int (x + 1)\, dx = 1 + x + x^2/2$

$y^{(2)} = 1 + \int [\, x + 1 + x + x^2/2 \,]\, dx = 1 + x + x^2 + x^3/6$

$y^{(3)} = 1 + \int [\, x + 1 + x + x^2 + x^3/6 \,]\, dx = 1 + x + x^2 + x^3/3 + x^4/24$

At $x = 0.1$: $y^{(3)} = 1 + 0.1 + 0.01 + 0.000333 + 0.0000042 = 1.11034$

Example 4

Find $y(0.2)$ from $dy/dx = x^2 + y^2$, $y(0) = 0$ (Picard).

$y^{(1)} = \int x^2\, dx = x^3/3$

$y^{(2)} = \int [\, x^2 + x^6/9 \,]\, dx = x^3/3 + x^7/63$

$y(0.2) \approx (0.2)^3/3 + (0.2)^7/63 = 0.002667 + 0.0000002 = 0.002667$

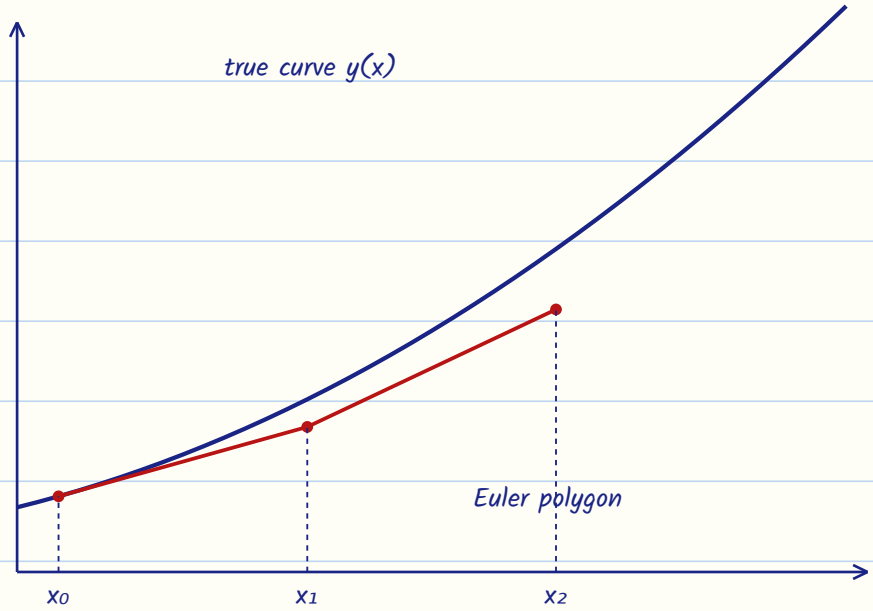
Remarks

- Picard's method gives y as a **function of x**; but integration becomes very difficult for complex $f(x, y)$.
- It is mostly used to prove existence & uniqueness theorem.

4. Euler's Method

Idea : in a small interval h, the curve is replaced by its tangent line at (x_n, y_n).

$$y_{n+1} = y_n + h\, f(x_n, y_n)\ ,\ x_{n+1} = x_n + h$$



$y_1 = y_0 + h\, f(x_0, y_0)$
step h along tangent
error grows at each step
(local error ~ h^2)

It is the simplest, first order method (error per step is of order h^2) but not very accurate.

Example 5

Solve $dy/dx = x + y$, $y(0) = 1$ with $h = 0.1$ for $x = 0.1, 0.2, 0.3$.

n	x_n	y_n	$f(x_n, y_n) = x_n + y_n$	$y_{n+1} = y_n + 0.1\, f$
0	0.0	1.0	1.0	1.1
1	0.1	1.1	1.2	1.22
2	0.2	1.22	1.42	1.362

So $y(0.1) = 1.1$, $y(0.2) = 1.22$, $y(0.3) = \mathbf{1.362}$. Exact value $y(0.3) = 1.39972$, so error = 0.0377 (large).

Demerits

- Low accuracy, needs very small h; errors accumulate; so we improve it (modified Euler).

5. Modified Euler's Method (Heun's Method)

Average the slopes at both ends of interval. First **predict** with Euler, then **correct** :

$$\text{Predictor} : y_{n+1}^{(0)} = y_n + h\, f(x_n, y_n)$$

$$\text{Corrector} : y_{n+1}^{(1)} = y_n + (h/2)\,[\, f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(0)})\,]$$

The corrector can be repeated till two successive values agree.

Example 6

Same problem $dy/dx = x + y$, $y(0) = 1$, $h = 0.1$.

Step 1 : $y^{(0)}(0.1) = 1 + 0.1(1) = 1.1$. Corrector : $y(0.1) = 1 + 0.05\,[\,1 + (0.1 + 1.1)\,] = 1 + 0.05(2.2) =$ **1.11**

Step 2 : $y^{(0)}(0.2) = 1.11 + 0.1(1.21) = 1.231$. $y(0.2) = 1.11 + 0.05\,[\,1.21 + (0.2 + 1.231)\,] =$ **1.24205**

Step 3 : $y^{(0)}(0.3) = 1.24205 + 0.1(1.44205) = 1.38626$. $y(0.3) = 1.24205 + 0.05\,[\,1.44205 + 1.68626\,] =$ **1.39847**

Much closer to exact 1.39972 than Euler's 1.362.

Example 7

$dy/dx = x + y^2$, $y(0) = 1$, $h = 0.1$. Predictor $y^{(0)}(0.1) = 1 + 0.1(1) = 1.1$. Corrector :

$y(0.1) = 1 + 0.05\,[\,1 + (0.1 + 1.21)\,] = 1 + 0.05(2.31) =$ **1.1155**

(Taylor gave 1.1165)

6. Runge-Kutta Methods

Give high accuracy **without** computing higher derivatives. Agree with Taylor's series up to h^p term (p = order). We need $f(x, y)$ only.

6.1 Second order R-K method

$$k_1 = h\, f(x_n, y_n)\ ,\ k_2 = h\, f(x_n + h, y_n + k_1)$$

$$y_{n+1} = y_n + \tfrac{1}{2}(k_1 + k_2)$$

(identical to modified Euler's method)

Example 8

$dy/dx = x + y$, $y(0) = 1$, $h = 0.1$. $k_1 = 0.1(1) = 0.1$; $k_2 = 0.1\, f(0.1, 1.1) = 0.1(1.2) = 0.12$

$y(0.1) = 1 + \tfrac{1}{2}(0.1 + 0.12) =$ **1.11**

6.2 Fourth order R-K method (most used)

$$k_1 = h\, f(x_n, y_n)\ ,\ k_2 = h\, f(x_n + h/2, y_n + k_1/2)$$

$$k_3 = h\, f(x_n + h/2, y_n + k_2/2)\ ,\ k_4 = h\, f(x_n + h, y_n + k_3)$$

$$y_{n+1} = y_n + (1/6)\,[\,k_1 + 2k_2 + 2k_3 + k_4\,]$$

Error of the 4th order R-K method is of order h^5 per step. Four function evaluations per step. Self-starting (needs only y_0).

Example 9

$dy/dx = x + y$, $y(0) = 1$, $h = 0.1$. Find $y(0.1)$.

$k_1 = 0.1(0 + 1) = 0.1$

$k_2 = 0.1\, f(0.05, 1.05) = 0.1(1.1) = 0.11$

$k_3 = 0.1\, f(0.05, 1.055) = 0.1(1.105) = 0.1105$

$k_4 = 0.1\, f(0.1, 1.1105) = 0.1(1.2105) = 0.12105$

$y(0.1) = 1 + (1/6)[0.1 + 0.22 + 0.221 + 0.12105] = 1 + 0.662105/6$

$= \mathbf{1.11034}$

Exact = 1.110342 : error less than 10^{-6} !

Example 10

$dy/dx = (y - x)/(y + x)$, $y(0) = 1$, $h = 0.2$. Find $y(0.2)$.

$k_1 = 0.2 \cdot (1 - 0)/(1 + 0) = 0.2$

$k_2 = 0.2\ f(0.1, 1.1) = 0.2\ (1.0/1.2) = 0.1667$

$k_3 = 0.2\ f(0.1, 1.0833) = 0.2\ (0.9833/1.1833) = 0.1662$

$k_4 = 0.2\ f(0.2, 1.1662) = 0.2\ (0.9662/1.3662) = 0.1414$

$y(0.2) = 1 + (1/6)[0.2 + 0.3334 + 0.3324 + 0.1414] = 1 +$

$1.0072/6 = \mathbf{1.1679}$

(Second order R-K for the same problem gives 1.1714.)

6A. More Solved Problems

Example 11 : Euler for a non-linear equation

$dy/dx = x + y^2$, $y(0) = 1$, $h = 0.1$. Find $y(0.1)$, $y(0.2)$, $y(0.3)$.

n	x_n	y_n	$f = x_n + y_n^2$	y_{n+1}
0	0.0	1.0	1.0	1.1
1	0.1	1.1	1.31	1.231
2	0.2	1.231	1.7155	1.4025

So $y(0.3) \approx 1.4025$ by Euler ; the modified Euler method gives 1.4820 for the same data, Taylor's series (4 terms) gives about 1.4825 : Euler is too low here.

Example 12 : Modified Euler, repeating the corrector

$dy/dx = x + y$, $y(0) = 1$, $h = 0.1$; find $y(0.1)$ accurate to 4 decimals.

$y^{(0)} = 1.1$. $y^{(1)} = 1 + 0.05\ [\ 1 + 1.2\ ] = 1.1100$

$y^{(2)} = 1 + 0.05\ [\ 1 + (0.1 + 1.11)\ ] = 1.1105$

$y^{(3)} = 1 + 0.05\ [\ 1 + (0.1 + 1.1105)\ ] = 1.11053$

Two successive values agree (1.1105), so **$y(0.1) = 1.1105$** (exact 1.1103).

Example 13 : R-K 4th order, second step

Continue Example 9. Now $x_1 = 0.1$, $y_1 = 1.11034$, $h = 0.1$. Find $y(0.2)$.

$k_1 = 0.1(0.1 + 1.11034) = 0.121034$

$k_2 = 0.1(0.15 + 1.17086) = 0.132086$

$k_3 = 0.1(0.15 + 1.17638) = 0.132638$

$k_4 = 0.1(0.2 + 1.24298) = 0.144298$

$y(0.2) = 1.11034 + (1/6)[0.121034 + 0.264172 + 0.265276 + 0.144298] = 1.11034 + 0.132464 =$ **1.24280**

(Exact value 1.242805 ✓)

Example 14 : Taylor's series in steps

$dy/dx = x - y^2$, $y(0) = 1$. Find $y(0.1)$ by Taylor's series (first 4 terms).

$y' = x - y^2 \to -1$; $y'' = 1 - 2yy' \to 1 + 2 = 3$; $y''' = -2(y'^2 + yy'') \to -2(1 + 3) = -8$

$y = 1 - x + 3x^2/2 - 8x^3/6 + \ldots$

$y(0.1) = 1 - 0.1 + 0.015 - 0.00133 = \mathbf{0.9137}$

Example 15 : Picard for another equation

$dy/dx = 1 + y^2$, $y(0) = 0$. (exact solution $y = \tan x$)

$y^{(1)} = \int (1 + 0)\, dx = x$

$y^{(2)} = \int (1 + x^2)\, dx = x + x^3/3$

$y^{(3)} = \int [1 + (x + x^3/3)^2]\, dx = x + x^3/3 + 2x^5/15 + x^7/63$

Compare with $\tan x = x + x^3/3 + 2x^5/15 + 17x^7/315 + \ldots$: first three terms already match.

Derivation of Euler's formula

From Taylor's series $y(x_0 + h) = y_0 + h\, y_0' + (h^2/2)\, y_0'' + \ldots$

Neglect h^2 and higher terms : $y_1 = y_0 + h\, f(x_0, y_0)$. So local truncation error is $O(h^2)$ and global error $O(h)$.

Derivation idea of modified Euler

Integrate $y' = f(x, y)$ from x_n to x_{n+1} and use the **trapezoidal rule** : $y_{n+1} - y_n = (h/2)[\, f(x_n, y_n) + f(x_{n+1}, y_{n+1})\,]$. Since y_{n+1} on the right side is unknown, estimate it by Euler's formula (predictor).

Points to remember while solving R-K problems

- Always compute k_1, k_2, k_3, k_4 **in order**; each uses the previous k.
- Keep 4 to 5 decimal places in the intermediate work.
- To go from x_0 to $x_0 + 2h$, do R-K twice, taking the answer of first step as the new y_0.
- If h is not given, choose $h = (x_final - x_0)/n$, e.g. $n = 1$ or 2.

7. Comparison of the Methods

For $y' = x + y$, $y(0) = 1$, exact $y(0.3) = 1.39972$.

Method	y(0.3)	Error
Euler	1.3620	0.0377
Modified Euler	1.3985	0.0012
R-K 2nd order (h = 0.1)	similar to mod. Euler	small
R-K 4th order	1.39972	$< 10^{-5}$
Method	Order	Remark
Euler	1	simple, poor accuracy
Modified Euler / RK2	2	predictor-corrector
RK4	4	very accurate, popular
Taylor	any	needs derivatives
Picard	any	gives series in x

8. Quick Revision

- Taylor : $y = y_0 + (x-x_0)y_0' + (x-x_0)^2 y_0''/2! + \dots$
- Picard : $y^{(n)} = y_0 + \int f(x, y^{(n-1)})\, dx$
- Euler : $y_{n+1} = y_n + h\, f(x_n, y_n)$
- Modified Euler : $y_{n+1} = y_n + h/2\ [\ f(x_n, y_n) + f(x_{n+1}, y_{n+1}^{(0)})\ ]$
- RK4 : $y_{n+1} = y_n + (k_1 + 2k_2 + 2k_3 + k_4)/6$

Exam questions

- Using Taylor's series find $y(0.1)$ for $y' = x^2 + y^2$, $y(0) = 1$.
- Find $y(0.2)$ by Picard's method for $y' = x + y$, $y(0) = 1$.
- Solve $y' = x - y^2$, $y(0) = 1$ for $y(0.2)$ by modified Euler, $h = 0.1$.
- Use R-K method of 4th order to find $y(0.2)$ with $h = 0.1$ for $y' = x + y$, $y(0) = 1$ (answer 1.24281).

NUMERICAL & STATISTICAL METHODS

Subject Code : 23AS4406

UNIT - IV

Estimation and Testing of Hypothesis, Large Sample Tests

Syllabus: Estimation - parameters, statistics, sampling distribution, point estimation, formulation of null and alternative hypothesis, critical and acceptance regions, level of significance, two types of errors and power of the test. Large sample tests: single proportion, difference of proportions, single mean, difference of means. Confidence interval for parameters in one sample and two sample problems.

I. Basic Terms

- **Population (universe)** : the complete collection of items under study. **Size N**.
- **Sample** : a part of the population selected for study. **Size n**. ($n \geq 30$: large sample, $n < 30$: small sample)
- **Parameter** : a numerical characteristic of the population, e.g. mean μ, variance σ^2, proportion P.
- **Statistic** : a numerical characteristic of the sample, e.g. $\bar{x}$, s^2, p. A statistic is a random variable.
- **Sampling** : random sampling with replacement (SRSWR) or without replacement (SRSWOR).

Sampling distribution : the probability distribution of a statistic computed from all possible samples of the same size n drawn from a population.

Standard error (S.E.) : standard deviation of the sampling distribution of a statistic. It measures the reliability of the statistic.

Important results

Statistic	Mean	Standard error
Sample mean $\bar{x}$	μ	$\sigma/\sqrt{n}$
Sample proportion p	P	$\sqrt{(PQ/n)}$
Difference of means $\bar{x}_1 - \bar{x}_2$	$\mu_1 - \mu_2$	$\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$
Difference of proportions $p_1 - p_2$	$P_1 - P_2$	$\sqrt{(P_1Q_1/n_1 + P_2Q_2/n_2)}$

($Q = 1 - P$). If population is finite and sampling is without replacement multiply S.E. by $\sqrt{[(N - n)/(N - 1)]}$.

Central Limit Theorem (CLT)

If a random sample of size n is drawn from **any** population with mean μ and variance σ^2, then for large n ($n \geq 30$) the sampling distribution of $\bar{x}$ is approximately **normal** $N(\mu, \sigma^2/n)$. Hence $\mathbf{Z = (\bar{x} - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)}$.

2. Estimation

Estimation is the process of finding the value of a population parameter using sample data. The rule/formula used is called an

estimator, its value for a given sample is an **estimate**.

2.1 Point estimation

A **single value** is given as estimate of the parameter. Example : $\bar{x}$ is a point estimate of μ ; s^2 estimates σ^2 ; p estimates P.

Characteristics of a good estimator

- **Unbiased** : $E(T) = \theta$ (e.g. $E(\bar{x}) = \mu$). Sample variance with divisor $(n - 1)$ is unbiased for σ^2.
- **Consistent** : $T \to \theta$ as $n \to \infty$ (variance of T tends to 0).
- **Efficient** : among unbiased estimators, the one with minimum variance.
- **Sufficient** : T uses all information about θ contained in the sample.

2.2 Interval estimation

An interval (L, U) such that $P(L < \theta < U) = 1 - \alpha$. It is called a **$(1 - \alpha) \times 100$ % confidence interval**; L and U are **confidence limits**. Example : 95 % C.I. means 95 times out of 100 the interval contains the true parameter.

Confidence level	90 %	95 %	98 %	99 %
Critical value z	1.645	1.96	2.33	2.58

3. Testing of Hypothesis : Concepts

A **statistical hypothesis** is a statement about a parameter / distribution of the population which is tested on the basis of

sample.

3.1 Null and alternative hypothesis

- **Null hypothesis H_0** : hypothesis of "no difference", tested for rejection. Example $H_0 : \mu = \mu_0$.
- **Alternative hypothesis H_1** : what we accept if H_0 is rejected. Examples $H_1 : \mu \neq \mu_0$ (two-tailed), $H_1 : \mu > \mu_0$ (right-tailed), $H_1 : \mu < \mu_0$ (left-tailed).

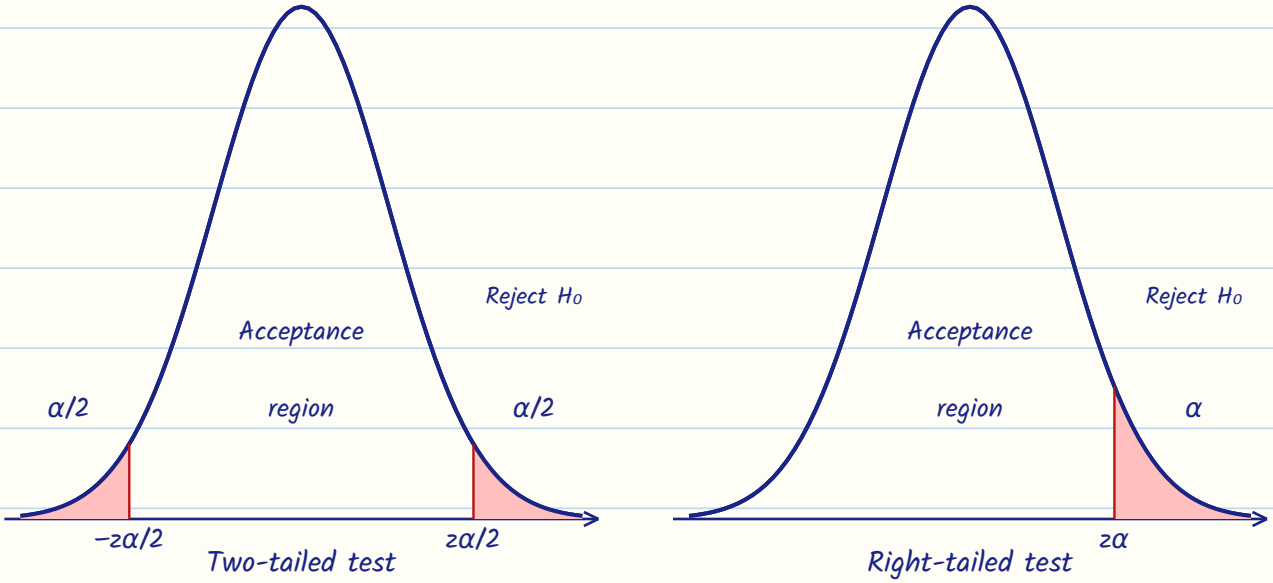
3.2 Test statistic, critical region

- **Test statistic** : a function of the sample values used to decide (like z).
- **Critical (rejection) region** : set of values of test statistic for which H_0 is rejected.
- **Acceptance region** : the remaining values where H_0 is accepted.
- **Critical value** : the boundary between the two regions.

3.3 Level of significance (α)

Probability of rejecting H_0 when it is true. Usually **5 %** or **1 %** (sometimes 10 %).

Level of significance	1 %	5 %	10 %						
Two-tailed test		z	= 2.58		z	= 1.96		z	= 1.645
Right-tailed test	z = 2.33	z = 1.645	z = 1.28						
Left-tailed test	z = −2.33	z = −1.645	z = −1.28						



3.4 Two types of errors

Decision / Truth	H_0 true	H_0 false
Reject H_0	**Type I error** (α)	Correct (power $1-\beta$)
Accept H_0	Correct	**Type II error** (β)

Type I error : rejecting H_0 when it is true, $P = \alpha$ (producer's risk). **Type II error** : accepting H_0 when it is false, $P = \beta$ (consumer's risk). **Power of the test** $= 1 - \beta$ = probability of rejecting H_0 when H_1 is true.

For a fixed sample size, decreasing α increases β. Increasing n reduces both.

3.5 Procedure of testing

- 1. Set up H_0 and H_1 (decides one or two-tailed).
- 2. Choose the level of significance α.
- 3. Select the suitable test statistic and compute its value.
- 4. Find the critical value from the table.
- 5. Decision : if |computed| > critical value, **reject** H_0, otherwise

accept H_0 (no evidence against it).

- 6. Write the conclusion in words.

4. Large Sample Tests ($n \geq 30$) : Z-tests

By CLT the sampling distributions are normal, so the test statistic

Z = (statistic − E(statistic)) / S.E.(statistic) ~ $N(0, 1)$.

4.1 Test for single mean

$H_0 : \mu = \mu_0$. \$ $Z = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$ (if σ unknown use s in place of σ)

Example 1

A sample of 400 items has mean 99. Population mean claimed is 100 with S.D. 8. Test at 5 % level.

$H_0 : \mu = 100$; $H_1 : \mu \neq 100$ (two-tailed).

$Z = (99 - 100)/(8/\sqrt{400}) = -1/0.4 =$ **−2.5**

$|Z| = 2.5 > 1.96$, so H_0 is **rejected** at 5 % level. (At 1 % level the critical value is 2.58, so H_0 is accepted at 1 %.)

4.2 Test for single proportion

Sample proportion $p = x/n$, $H_0 : P = P_0$. \$ $Z = (p - P_0) / \sqrt{(P_0Q_0/n)}$

Example 2

In a sample of 400 articles, 30 are defective. Can we say the proportion of defectives in the lot is 5 % ? ($\alpha = 0.05$)

$H_0 : P = 0.05$; $H_1 : P \neq 0.05$. $p = 30/400 = 0.075$.

S.E. = $\sqrt{(0.05 \times 0.95/400)} = 0.01090$

$Z = (0.075 - 0.05)/0.01090 = \mathbf{2.29} > 1.96 \rightarrow H_0$ **rejected** : proportion of defectives is not 5 %. (At 1 % level 2.29 < 2.58, accepted.)

4.3 Test for difference of means

$H_0 : \mu_1 = \mu_2$. $$ $Z = (\bar{x}_1 - \bar{x}_2) / \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$

Example 3

Two samples of 1000 men each from two places have mean heights 67.5 and 68.0 inches with S.D. 2.5 and 2.8. Test whether the places differ in mean height (5 %).

$Z = (67.5 - 68.0)/\sqrt{(6.25/1000 + 7.84/1000)} = -0.5/0.1187 = \mathbf{-4.21}$

$|Z| = 4.21 > 1.96 \rightarrow H_0$ rejected. The mean heights differ **significantly**.

4.4 Test for difference of proportions

$H_0 : P_1 = P_2$. Use pooled proportion $\hat{p} = (x_1 + x_2)/(n_1 + n_2) = (n_1p_1 + n_2p_2)/(n_1 + n_2)$

$$Z = (p_1 - p_2) / \sqrt{[\ \hat{p}\ \hat{q}\ (1/n_1 + 1/n_2)\ ]}$$

Example 4

In a sample of 400 men of town A, 40 smoke ; in 600 men of town B, 30 smoke. Is there a difference in the proportion of smokers ? (5 %)

$p_1 = 40/400 = 0.10$, $p_2 = 30/600 = 0.05$, $\hat{p} = 70/1000 = 0.07$, $\hat{q} = 0.93$

S.E. = $\sqrt{[0.07 \times 0.93\ (1/400 + 1/600)]} = \sqrt{(0.0651 \times 0.004167)} =$

0.01647

$Z = 0.05/0.01647 = \mathbf{3.04} > 1.96 \rightarrow H_0$ rejected (difference is significant).

Summary of Z-tests

Test	H_0	Z statistic
Single mean	$\mu = \mu_0$	$(\bar{x} - \mu_0)/(\sigma/\sqrt{n})$
Single proportion	$P = P_0$	$(p - P_0)/\sqrt{(P_0Q_0/n)}$
Two means	$\mu_1 = \mu_2$	$(\bar{x}_1 - \bar{x}_2)/\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$
Two proportions	$P_1 = P_2$	$(p_1 - p_2)/\sqrt{(\hat{p}\hat{q}(1/n_1 + 1/n_2))}$

5. Confidence Intervals

5.1 One-sample problems

- Mean (σ known or n large) : $\bar{x} \pm z\ (\sigma/\sqrt{n})$
- Proportion : $p \pm z\ \sqrt{(pq/n)}$

Here z = 1.96 for 95 %, 2.58 for 99 %.

Example 5

A sample of 100 has mean 65, population S.D. = 5. Find the 95 % confidence interval for μ.

$\bar{x} \pm 1.96\ \sigma/\sqrt{n} = 65 \pm 1.96\ (5/10) = 65 \pm 0.98 \rightarrow$ **(64.02 , 65.98)**

For 99 % : $65 \pm 2.58(0.5) = (63.71 , 66.29)$. Higher confidence gives a wider interval.

Example 6 (proportion)

In a sample of 400 voters 240 favour a candidate. 95 % C.I. for the true proportion :

$p = 0.6$, S.E. $= \sqrt{(0.6 \times 0.4/400)} = 0.0245$. Interval $= 0.6 \pm 1.96(0.0245) = 0.6 \pm 0.048 \rightarrow$ **(0.552 , 0.648)**

5.2 Two-sample problems

- Difference of means : $(\bar{x}_1 - \bar{x}_2) \pm z\sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$
- Difference of proportions : $(p_1 - p_2) \pm z\sqrt{(p_1q_1/n_1 + p_2q_2/n_2)}$

Example 7

Using the data of Example 3, 95 % C.I. for $\mu_1 - \mu_2$:

$(67.5 - 68.0) \pm 1.96\ (0.1187) = -0.5 \pm 0.233 \rightarrow$ **(−0.733 , −0.267)**. Zero is not inside, which agrees with rejecting H_0.

Sample size for estimating the mean

For error not more than E with confidence $(1 - \alpha)$: $n = (z\sigma / E)^2$

Example : $\sigma = 5$, $E = 1$, 95 % $\rightarrow n = (1.96 \times 5/1)^2 = 96.04 \approx$ **97**.

5A. More Solved Problems

Example 8 : single mean, σ unknown (n ≥ 30)

A sample of 64 students has mean marks 52 and S.D. 8. Test whether the population mean is 50 (5 %).

$H_0 : \mu = 50$; $H_1 : \mu \neq 50$. $z = (52 - 50)/(8/\sqrt{64}) = 2/1 =$ **2.0**

$2.0 > 1.96 \rightarrow H_0$ rejected at 5 % (just significant). At 1 % level it

would be accepted.

Example 9 : one-tailed (left) test

A bulb maker claims average life 600 hours. A sample of 100 bulbs has mean 580 hours with S.D. 50. Is the claim valid or is the true life less ? (5 %)

$H_0 : \mu = 600$; $H_1 : \mu < 600$ (left-tailed). $Z = (580 - 600)/(50/10) = \mathbf{-4.0}$

Critical value for left tail at 5 % is −1.645. Since $-4.0 < -1.645$, H_0 **rejected** : life of bulbs is less than 600 hours.

Example 10 : one-tailed test for proportion

A medicine is said to cure 90 % patients. In a sample of 500, 440 cured ($p = 0.88$). Is the claim justified ? (1 %)

$H_0 : P = 0.90$; $H_1 : P < 0.90$. S.E. $= \sqrt{(0.9 \times 0.1/500)} = 0.01342$

$Z = (0.88 - 0.90)/0.01342 = \mathbf{-1.49}$. Critical value −2.33. Since $|-1.49| < 2.33 \rightarrow H_0$ **accepted** : claim can be accepted.

Example 11 : two means with sample S.D.

Sample 1 : $n_1 = 50$, $\bar{x}_1 = 82$, $s_1 = 6$. Sample 2 : $n_2 = 60$, $\bar{x}_2 = 80$, $s_2 = 7$. Test $H_0 : \mu_1 = \mu_2$ at 5 %.

$Z = (82 - 80)/\sqrt{(36/50 + 49/60)} = 2/\sqrt{(0.72 + 0.8167)} = 2/1.2397 = \mathbf{1.61}$

$1.61 < 1.96 \rightarrow H_0$ accepted : no significant difference.

Example 12 : confidence interval for difference of

From Example 4 : $p_1 = 0.10$ ($n_1 = 400$), $p_2 = 0.05$ ($n_2 = 600$).

S.E. $= \sqrt{[0.1(0.9)/400 + 0.05(0.95)/600]} = \sqrt{(0.000225 +}$

$0.0000792) = 0.01744$

95 % C.I. for $P_1 - P_2 = 0.05 \pm 1.96(0.01744) = 0.05 \pm 0.0342 \rightarrow$ **(0.016 , 0.084)**

Example 13 : Type I and Type II errors numerically

Test $H_0 : \mu = 50$ against $H_1 : \mu = 56$ using $n = 25$, $\sigma = 10$, rule : reject H_0 if $\bar{x} > 54$.

S.E. $= 10/\sqrt{25} = 2$.

$\alpha = P(\bar{x} > 54 \mid \mu = 50) = P(Z > (54 - 50)/2 = 2) = 1 - 0.9772$ $=$ **0.0228**

$\beta = P(\bar{x} \leq 54 \mid \mu = 56) = P(Z \leq (54 - 56)/2 = -1) =$ **0.1587**

Power $= 1 - \beta = 0.8413$.

Relation between CI and test

A two-sided test of $H_0 : \theta = \theta_0$ at level α rejects H_0 exactly when θ_0 lies **outside** the $(1 - \alpha)$ confidence interval. For Example 5, 66 is outside (64.02, 65.98), so $H_0 : \mu = 66$ would be rejected at 5 %.

Steps to avoid mistakes

- Check whether σ or only s is given ; for $n \geq 30$ both work with Z.
- Read the question for words like "less than", "greater than" (one-tailed) or "differ", "not equal" (two-tailed).
- For two proportions always use the **pooled** $\hat{p}$ in the test, but p_1q_1 and p_2q_2 in the confidence interval.
- Write conclusion in words : "there is a significant difference" or "no evidence against H_0".

p-value idea (extra)

p-value = probability of getting a result at least as extreme as observed, if H_0 is true. Small p-value ($< \alpha$) means reject H_0. For $Z = 2.5$ two-tailed $p = 2 \times 0.0062 = 0.0124$ which is < 0.05 : reject at 5 %.

6. Quick Revision

- Statistic vs parameter ; S.E. of $\bar{x} = \sigma/\sqrt{n}$; S.E. of $p = \sqrt{(PQ/n)}$.
- Type I error α, Type II error β, power $= 1 - \beta$.
- Critical z : 5 % two-tailed 1.96, one-tailed 1.645 ; 1 % two-tailed 2.58, one-tailed 2.33.
- Reject H_0 if $|Z|$ > critical value.
- C.I. : estimate $\pm$ z $\times$ S.E.

Exam questions

- Define null hypothesis, level of significance, type I and type II errors.
- A coin is tossed 400 times, head appears 216 times. Is the coin unbiased ? (z = 1.6, accept at 5 %).
- Test equality of means of two large samples (given n, $\bar{x}$, s).
- Find 95 % and 99 % confidence limits for the mean from a large sample.

NUMERICAL & STATISTICAL METHODS

Subject Code : 23AS4406

UNIT - V

Small Sample Tests

Syllabus: Student t-distribution (test for single mean, two means and paired t-test), testing of equality of variances (F-test), chi-square test for goodness of fit, chi-square test for independence of attributes.

I. Small Sample Tests : Introduction

When the sample size **$n < 30$**, the normal (Z) test is not reliable because the sampling distribution of the statistic is not normal. We use **exact sampling distributions** : t, F and χ^2 distributions. These need the assumption that the **parent population is normal**.

Test	Used for	Distribution
t-test	mean(s) of small samples	Student's t, df = ν
F-test	equality of two variances	Snedecor's F, (ν_1, ν_2)
χ^2 test	goodness of fit, independence	Chi-square, df = ν

Degrees of freedom (df) : number of independent observations available for estimating a quantity = (number of observations) − (number of constraints / estimated parameters).

2. Student's t-distribution

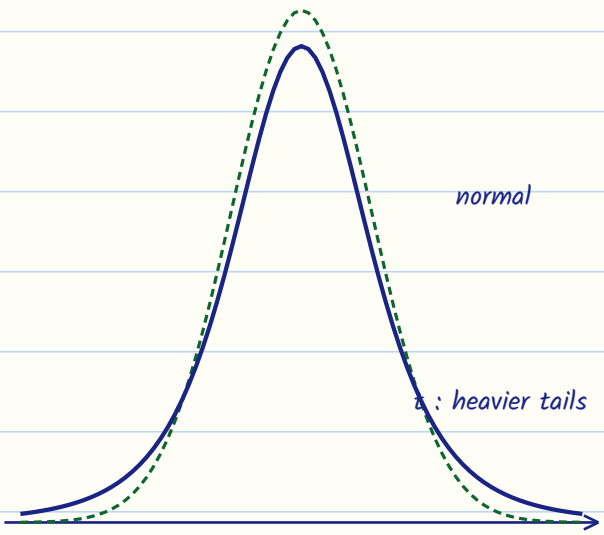
Let $x_1, x_2 \ldots x_n$ be a random sample from $N(\mu, \sigma^2)$. Define

$$t = (\bar{x} - \mu) / (s/\sqrt{n}) \ , \ s^2 = \Sigma (x_i - \bar{x})^2 / (n - 1)$$

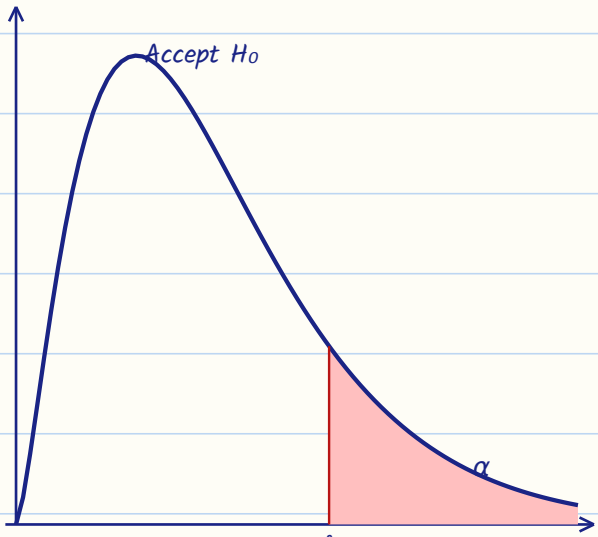
t follows **Student's t-distribution** with $\nu = n - 1$ degrees of freedom. (W. S. Gosset, pen name "Student").

Properties of the t-distribution

- Bell shaped and **symmetric about t = 0** ; mean = 0 ; variance = $\nu/(\nu - 2)$ for $\nu > 2$.
- Flatter than normal curve with **heavier tails** ; as $\nu \to \infty$ it tends to $N(0, 1)$.
- It depends only on ν (not on μ, σ).
- Total area under the curve = 1 ; critical values are read from t-table for (ν, α).



t-distribution (n small)



χ^2 distribution (skewed)

Selected t-table values (two-tailed)

df	5	7	9	11	15	20	∞
α = 0.05	2.571	2.365	2.262	2.201	2.131	2.086	1.960
α = 0.01	4.032	3.499	3.250	3.106	2.947	2.845	2.576

Applications of t

- Test for a single mean ; test for difference of two means (independent) ; paired t-test ; confidence limits for mean ; test for correlation coefficient.

3. t-test for Single Mean

$H_0 : \mu = \mu_0$. $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$ with $n - 1$ df, where $s = \sqrt{[\Sigma(x - \bar{x})^2/(n - 1)]}$.

Procedure

- $H_0 : \mu = \mu_0$; $H_1 : \mu \neq \mu_0$ (or one-sided). Select α.
- Compute $\bar{x}$, s and then t.
- Compare $|t|$ with the table value $t_\alpha(n - 1)$. If $|t|$ > table value → reject H_0.

Example 1

Ten individuals have the heights 70, 72, 68, 74, 66, 71, 69, 73, 67, 70 (inches). Is it reasonable to believe that the mean height of the population is 68 inches ? (5 %)

$\bar{x} = 700/10 = 70$. Deviations $(x - \bar{x})$: 0, 2, −2, 4, −4, 1, −1, 3, −3, 0.

$\Sigma(x - \bar{x})^2 = 0 + 4 + 4 + 16 + 16 + 1 + 1 + 9 + 9 + 0 = 60$; $s^2 = 60/9 = 6.667$; $s = 2.582$

$t = (70 - 68)/(2.582/\sqrt{10}) = 2/0.8165 = \mathbf{2.449}$

$df = 9$, table $t_{0.05} = 2.262$. Since $2.449 > 2.262$, H_0 is **rejected** : the mean height is not 68 inches.

Confidence limits for μ (small sample)

$\bar{x} \pm t_\alpha \cdot s/\sqrt{n}$. For Example 1 (95 %) : $70 \pm 2.262\ (0.8165) = 70 \pm 1.85 \rightarrow$ **(68.15 , 71.85)**. As 68 is outside, H_0 rejected ✓.

4. t-test for Difference of Two Means (independent

Two independent small samples from normal populations with **equal variances** $\sigma_1^2 = \sigma_2^2$. $H_0 : \mu_1 = \mu_2$.

$$t = (\bar{x}_1 - \bar{x}_2) \,/\, [\, s\sqrt{(1/n_1 + 1/n_2)}\,] \text{ with } n_1 + n_2 - 2 \text{ df}$$

$$s^2 = [\, \Sigma(x_1 - \bar{x}_1)^2 + \Sigma(x_2 - \bar{x}_2)^2\,] \,/\, (n_1 + n_2 - 2) \text{ (pooled variance)}$$

Example 2

Sample 1 : 20, 16, 26, 27, 23, 22 ($n_1 = 6$). Sample 2 : 27, 33, 42, 35, 32, 34, 38 ($n_2 = 7$). Test whether the means differ (5 %).

$\bar{x}_1 = 134/6 = 22.33$; $\Sigma(x_1 - \bar{x}_1)^2 = 81.33$

$\bar{x}_2 = 241/7 = 34.43$; $\Sigma(x_2 - \bar{x}_2)^2 = 133.71$

$s^2 = (81.33 + 133.71)/11 = 19.55$, $s = 4.421$

S.E. $= 4.421\sqrt{(1/6 + 1/7)} = 4.421 \times 0.5563 = 2.460$

$t = (22.33 - 34.43)/2.460 = \mathbf{-4.92}$. $df = 11$, $t_{0.05} = 2.201$.

$|t| = 4.92 > 2.201 \rightarrow H_0$ **rejected** : the two means differ

significantly.

> Before using this test one should check that $\sigma_1^2 = \sigma_2^2$ using the **F-test** (next section).

5. Paired t-test

Used when the observations are in **pairs on the same units** (before & after treatment, two methods on same subjects). Let d_i = difference of each pair. $H_0 : \mu_d = 0$.

$t = \bar{d} / (s_d/\sqrt{n})$, $\bar{d} = \Sigma d/n$, $s_d^2 = \Sigma(d - \bar{d})^2/(n - 1)$, $df = n - 1$

Example 3

Marks of 8 students before and after coaching :

Before	8	10	12	15	11	9	14	13
After	10	11	14	17	12	10	13	16
d = After − Before	2	1	2	2	1	1	−1	3

$\Sigma d = 11$, $\bar{d} = 1.375$; $\Sigma d^2 = 25$; $s_d^2 = [25 - 8(1.375)^2]/7 = (25 - 15.125)/7 = 1.4107$; $s_d = 1.188$

$t = 1.375/(1.188/\sqrt{8}) = 1.375/0.420 = \mathbf{3.27}$, $df = 7$, $t_{0.05} = 2.365$.

$3.27 > 2.365 \rightarrow H_0$ rejected : coaching **has improved** the marks (significantly).

Difference between independent and paired t

Independent t	Paired t
Two separate groups	Same group measured twice
$df = n_1 + n_2 - 2$	$df = n - 1$
Uses pooled variance	Uses variance of differences d

6. F-test : Equality of Two Variances

F-distribution (Snedecor) : ratio of two independent sample variances from normal populations.

$$F = s_1^2 / s_2^2 \text{ with } (\nu_1, \nu_2) = (n_1 - 1 , n_2 - 1)$$

(take the larger variance in the numerator, so that $F \geq 1$).

$H_0 : \sigma_1^2 = \sigma_2^2$; $H_1 : \sigma_1^2 \neq \sigma_2^2$ (or $>$). Reject H_0 if $F > F_{_\alpha}(\nu_1, \nu_2)$ from the F-table.

Properties of F

- Positively skewed, never negative ; shape depends on (ν_1, ν_2) ; mean $= \nu_2/(\nu_2 - 2)$; $F(\nu_1, \nu_2) = 1/F(\nu_2, \nu_1)$.

Example 4

Use the two samples of Example 2 to test $\sigma_1^2 = \sigma_2^2$ at 5 %.

$s_1^2 = 81.33/5 = 16.27$ $(n_1 = 6)$; $s_2^2 = 133.71/6 = 22.29$ $(n_2 = 7)$.

$F = 22.29/16.27 =$ **1.37** with (6, 5) df. Table $F_{0.05}(6, 5) = 4.95$.

$1.37 < 4.95 \rightarrow H_0$ accepted : the variances are equal, so the pooled t-test in Example 2 was valid ✓.

Example 5

Two samples : $n_1 = 9$, $s_1^2 = 40$; $n_2 = 8$, $s_2^2 = 10$. Is $\sigma_1^2 > \sigma_2^2$? (5 %, one-tailed)

$F = 40/10 = 4.0$ with (8, 7) df ; $F_{0.05}(8, 7) = 3.73$. Since $4.0 > 3.73$, **H_0 rejected** : σ_1^2 is greater.

7. Chi-Square (χ^2) Test

If O = observed frequency and E = expected frequency in each class then

$$\chi^2 = \Sigma (O - E)^2 / E \text{ with } \nu \text{ degrees of freedom}$$

Properties of χ^2 distribution

- Values from 0 to ∞ ; **positively skewed** ; mean = ν, variance = 2ν.
- Sum of two independent χ^2 variables is again χ^2 with df added.
- For large ν, χ^2 tends to normal.

Conditions for validity

- Total frequency N should be large ($N > 50$) ; each expected frequency ≥ 5 (otherwise pool neighbouring classes).
- Observations are independent ; expected frequencies sum to total of observed.

df	1	2	3	4	5	6
$\chi^2_{0.05}$	3.841	5.991	7.815	9.488	11.070	12.592
$\chi^2_{0.01}$	6.635	9.210	11.345	13.277	15.086	16.812

7.1 χ^2 test for goodness of fit

H_0 : the data follow the specified (theoretical) distribution. df = (number of classes − 1 − number of parameters estimated from data).

Example 6

A die is thrown 60 times. Observed : 1→8, 2→12, 3→9, 4→11, 5→14, 6→6. Is the die fair ? (5 %)

H_0 : die is fair, E = 60/6 = 10 for each face.

Face	1	2	3	4	5	6
O	8	12	9	11	14	6
E	10	10	10	10	10	10
$(O-E)^2/E$	0.4	0.4	0.1	0.1	1.6	1.6

$\chi^2 = 4.2$, df = 5 , table value 11.07. Since $4.2 < 11.07 \rightarrow H_0$ **accepted** : die is fair.

Example 7 : genetics ratio 9 : 3 : 3 : 1

Observed 315, 108, 101, 32 (total 556). Expected = 556 × (9, 3, 3, 1)/16 = 312.75, 104.25, 104.25, 34.75.

$\chi^2 = (2.25^2/312.75) + (3.75^2/104.25) + (3.25^2/104.25) + (2.75^2/34.75) = 0.016 + 0.135 + 0.101 + 0.218 =$ **0.470**

df = 3, table 7.815. 0.47 < 7.815 → accepted : data agree with the ratio 9 : 3 : 3 : 1.

7.2 χ^2 test for independence of attributes

Data are arranged in a **contingency table** (r rows × c columns). H_0 : the two attributes A and B are independent.

Expected frequency E = (row total × column total) / grand total

$df = (r - 1)(c - 1)$. Compute $\chi^2 = \Sigma(O - E)^2/E$.

Example 8 (2 × 2)

Smoking and lung disease :

	Disease	No disease	Total
Smokers	30	20	50
Non-smokers	10	40	50
Total	40	60	100

$E(30) = 50 \times 40/100 = 20$, $E(20) = 30$, $E(10) = 20$, $E(40) = 30$.

$\chi^2 = 10^2/20 + 10^2/30 + 10^2/20 + 10^2/30 = 5 + 3.33 + 5 + 3.33 =$ **16.67**

$df = (2 - 1)(2 - 1) = 1$, table value 3.841. $16.67 > 3.841 \rightarrow H_0$ rejected : disease and smoking are **dependent (associated)**.

Short cut for 2 × 2 table with cells a b / c d : $\chi^2 = N(ad - bc)^2 / [(a + b)(c + d)(a + c)(b + d)]$ Check : $100(1200 - 200)^2/(50 \cdot 50 \cdot 40 \cdot 60) = 16.67$ ✓

Yates' correction (for 2 × 2 with small cell, < 5)

Replace $|O - E|$ by $|O - E| - ½$ in each term ; $\chi^2 = N(|ad - bc| - N/2)^2 / [(a+b)(c+d)(a+c)(b+d)]$.

Example 9 (3 × 2)

Education level vs opinion (count of people) :

	Agree	Disagree	Total
Low	20	30	50
Medium	40	30	70
High	40	40	80
Total	100	100	200

Expected : Low 25, 25 ; Medium 35, 35 ; High 40, 40.

$\chi^2 = 25/25 + 25/25 + 25/35 + 25/35 + 0 + 0 = 1 + 1 + 0.714 + 0.714 =$ **3.43**

$df = (3 - 1)(2 - 1) = 2$, table 5.991. $3.43 < 5.991 \rightarrow H_0$ accepted : opinion is independent of education level.

7A. More Solved Problems

Example 10 : one-tailed t-test

A sample of 16 bulbs has mean life 53 hours and $s = 6$. Is the mean life greater than 50 hours ? (5 %)

$H_0 : \mu = 50$; $H_1 : \mu > 50$ (right-tailed). $t = (53 - 50)/(6/\sqrt{16}) = 3/1.5 =$ **2.0**

$df = 15$; one-tailed 5 % critical value = 1.753. Since $2.0 > 1.753 \rightarrow H_0$ **rejected** : mean life is greater than 50.

> For a **one-tailed** test at α use the table value of the two-tailed column 2α (i.e. the 10 % two-tailed column for 5 % one-tailed).

Example 11 : paired t-test (blood pressure)

Six patients : BP before 150, 160, 155, 170, 165, 158 ; after treatment 144, 155, 153, 160, 158, 155. Does the drug reduce BP ? (5 %)

d = before − after = 6, 5, 2, 10, 7, 3 ; $\bar{d} = 33/6 = 5.5$. Deviations : 0.5, −0.5, −3.5, 4.5, 1.5, −2.5 ; $\Sigma = 41.5$

$s_d^2 = 41.5/5 = 8.3$, $s_d = 2.881$. $t = 5.5/(2.881/\sqrt{6}) = 5.5/1.176 =$ **4.68**

$df = 5$, $t_{0.05}$ (two-tailed) = 2.571 ; 4.68 > 2.571 → H_0 rejected : the drug **does reduce** blood pressure.

Example 12 : F-test with sums of squares

Sample 1 : $n_1 = 10$, $\Sigma(x - \bar{x})^2 = 90 \rightarrow s_1^2 = 90/9 = 10$. Sample 2 : $n_2 = 12$, $\Sigma(y - \bar{y})^2 = 121 \rightarrow s_2^2 = 121/11 = 11$.

$F = 11/10 = 1.1$ with (11, 9) df ; $F_{0.05}(11, 9) = 3.10$. 1.1 < 3.10 → equal variances (accepted).

Example 13 : goodness of fit to a binomial law

4 coins are tossed 160 times. Number of heads 0, 1, 2, 3, 4 occurred 6, 32, 58, 50, 14 times. Is the coin fair ?

For a fair coin the frequencies are $160 \times (1, 4, 6, 4, 1)/16 = 10, 40, 60, 40, 10$.

Heads	0	1	2	3	4
O	6	32	58	50	14
E	10	40	60	40	10
$(O-E)^2/E$	1.6	1.6	0.067	2.5	1.6

$\chi^2 = 7.37$, $df = 4$, table 9.488. $7.37 < 9.488$ → accepted : the coins are fair (binomial fits).

Which test to use ? (decision guide)

- Comparing a **mean** with a standard value : $n \geq 30$ → Z ; $n < 30$ → t (single mean).
- Comparing **two means** : n large → Z ; small independent samples → t (pooled) ; same units twice → paired t.
- Comparing **two variances** → F.
- Comparing **observed with expected counts** or checking association between **attributes** → χ^2.

Z-test vs t-test

Z-test	t-test
$n \geq 30$ (large)	$n < 30$ (small)
Uses σ (or s)	Uses s with $(n - 1)$ df
Normal distribution	Student's t (heavier tails)
Critical value fixed 1.96	Depends on df (2.262 for 9 df)

Assumptions of small sample tests

- Population is (approximately) normal ; samples are random and independent (except paired).
- For the two-sample t-test, population variances are equal (check by F).
- χ^2 test is a **non-parametric** (distribution-free) test, no normality needed.

Relation between t, F and χ^2

- If t has ν df then t^2 follows $F(1, \nu)$.
- F is the ratio of two independent χ^2 variables each divided by its df : $F = (\chi_1^2/\nu_1)/(\chi_2^2/\nu_2)$.
- $(n-1)s^2/\sigma^2$ follows χ^2 with $(n-1)$ df.

8. Quick Revision

Test	Statistic	df
t (single mean)	$(\bar{x} - \mu)/(s/\sqrt{n})$	$n - 1$
t (two means)	$(\bar{x}_1 - \bar{x}_2)/(s\sqrt{(1/n_1 + 1/n_2)})$	$n_1 + n_2 - 2$
paired t	$\bar{d}/(s_d/\sqrt{n})$	$n - 1$
F	s_1^2/s_2^2 (larger on top)	$(n_1 - 1, n_2 - 1)$
χ^2	$\Sigma (O - E)^2/E$	$k - 1$ (fit), $(r-1)(c-1)$ (independence)

Exam questions

- State properties of t-distribution. Explain paired t-test with an example.
- Two samples of sizes 9 and 8 have sums of squared deviations 160 and 91 : test equality of variances using F-test ($F = 20.0/13.0 = 1.54$, accept).
- A die is thrown 120 times and the results are ... test goodness of fit using χ^2.
- Test independence of attributes for the 2 × 2 table given (use χ^2).